# CogMaPOI: A Cognition-Inspired Multi-Agent Framework for Continual Next POI Recommendation with Episodic Memory and Metacognitive Reflection

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*by*

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# Evaluation Sheet

**Title of the Project**:

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# Declaration

We, **Suguna Sri** and **Chilukala Amrutha**, with Roll No: **122CS0004, 122CS0065** hereby declare that the material presented in the Project Report titled **CogMaPOI: A Cognition-Inspired Multi-Agent Framework for Continual Next POI Recommendation with Episodic Memory and Metacognitive Reflection** represents original work carried out by us in the **Department of Computer Science and Engineering** at the **Indian Institute of Information Technology, Design and Manufacturing, Kurnool** during the years **2025 − 2026**.

With our signatures, we certify that:

- We have not manipulated any of the data or results.
- We have not committed any plagiarism of intellectual property. We have clearly indicated and referenced the contributions of others.
- We have explicitly acknowledged all collaborative research and discussions.
- We have understood that any false claim will result in severe disciplinary action.
- We have understood that the work may be screened for any form of academic misconduct.

Date: Students' Signatures

In my capacity as supervisor of the above-mentioned work, I certify that the work presented in this Report is carried out under my supervision, and is worthy of consideration for the requirements of B.Tech. Project work.

Advisor's Name: **Dr. K. E. Srinivasa Desikan** Advisor's Signature

# *Abstract*

Next point-of-interest recommendation has historically been treated as single-pass prediction: each output is produced independently, with no pathway for test-time observations to revise what follows. This design overlooks three faculties that shape human spatial decisions: stable long-term preferences, awareness of recent behavioural shifts, and the capacity to detect and correct one's own reasoning errors. CogMaPOI closes this gap through a five-agent LLM architecture that operationalises all three as separately maintained, mutually interacting components of a continual inference loop. A fixed identity prior encodes stable mobility preferences derived from historical check-ins; episodic memory accumulates natural-language summaries of test-session behaviour after each prediction step; and a metacognitive reflection record evaluates completed predictions, reinforcing strategies that succeeded and issuing targeted corrections when they did not. A Retrieval Agent assembles a user-specific candidate set before inference begins, and a Reranker integrates the full cognitive state at each step to produce the final ranked output. State evolution proceeds entirely through structured prompt updates with no parameter modifications at any stage, making the framework applicable on top of any instruction-tuned LLM without retraining.

Evaluated on three Foursquare datasets spanning New York City, Tokyo, and California against thirteen baselines, CogMaPOI achieves relative MRR improvements of up to 89.35% over the strongest per-metric baseline. The gains are consistently larger on ranking-sensitive metrics than on coverage metrics, reflecting that the cognitive loop improves the position of the correct candidate within the retrieved set rather than expanding retrieval coverage. Ablations confirm that memory and reflection make complementary, non-redundant contributions, with joint removal producing super-additive degradation. Even a 3B-parameter backbone outperforms the best static baseline on all metrics, establishing that the advantage is architectural rather than a consequence of model scale. These results position test-time cognitive adaptation as a viable, parameter-free alternative to fine-tuning for location prediction.

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# Abbreviations

| | |
|---|---|
| **POI** | **P**oint **o**f **I**nterest |
| **LLM** | **L**arge **L**anguage **M**odel |
| **MRR** | **M**ean **R**eciprocal **R**ank |
| **HR** | **H**it **R**ate |
| **NDCG** | **N**ormalised **D**iscounted **C**umulative **G**ain |
| **NYC** | **N**ew **Y**ork **C**ity (Foursquare dataset) |
| **TKY** | **T**o**k**yo (Foursquare dataset) |
| **CA** | **C**alifornia (Foursquare dataset) |
| **NF4** | **N**ormal**F**loat **4**-bit quantisation |
| **FP16** | **F**loating **P**oint **16**-bit precision |

*Dedicated to our parents, professors and friends, whose unwavering support, encouragement, and belief in us have been the foundation of all our achievements...*

# Chapter 1

# Introduction

## 1.1 Overview

Predicting where a person will go next is a problem with far-reaching practical value. Navigation assistants, transit planning tools, venue discovery platforms, and urban mobility analytics all depend on models capable of translating a user's movement history into reliable predictions about future destinations. As mobile devices have become ubiquitous and check-in data has grown in both volume and richness, accurate, personalised location prediction has emerged as a central challenge at the intersection of machine learning and spatial computing.

At the core of this problem is the concept of a point of interest (POI): any named, visitable location that a user can check into, such as a coffee shop, a transit hub, a park, or a museum. Given a sequence of past check-ins, the next-POI recommendation task asks a system to identify the single most probable destination a user will visit next. Although the task sounds straightforward, it carries considerable depth. A user's next destination depends not just on where they have been, but on who they are, what time it is, how their recent trajectory has unfolded, and whether their current behaviour departs from their established routines. Building a system that handles all these signals simultaneously, and does so without the ability to update its parameters during deployment, is the challenge that motivates this thesis.

## 1.2 Next-POI Recommendation

Formally, the next-POI recommendation task takes a user's accumulated visit history as a sequence of timestamped check-ins at named locations and ranks candidate destinations in decreasing order of predicted visit probability. The dominant framing is supervised sequence prediction: a model trained on historical trajectories is evaluated on held-out test sequences.

Over the past decade, this framing has attracted sustained research attention. Early work used recurrent networks to model temporal ordering in check-in sequences. These were followed by self-attention architectures adapted from natural language processing, graph-based encoders that capture spatial relationships between venues, and most recently by generative models that treat preference evolution as a continuous-time process. Every new family of methods has improved benchmark accuracy, yet all share the same structural constraint: once trained, their parameters are frozen. No observation made during evaluation can alter the model's behaviour. If a user's habits shift between training and evaluation, the model has no mechanism for incorporating that shift.

## 1.3 Large Language Models and Multi-Agent Frameworks

The arrival of large language models (LLMs) opened new possibilities for the recommendation problem. LLMs pre-trained on large text corpora with instruction-following objectives can interpret natural-language descriptions of user behaviour, reason about temporal and categorical patterns, and produce ranked outputs without task-specific fine-tuning. Applied to next-POI recommendation, these methods convert check-in histories into structured prompts and leverage the model's pre-existing semantic knowledge to reason about venue preferences and spatial context.

Multi-agent frameworks extend this paradigm by distributing a complex task across several specialised components, each responsible for a distinct sub-problem. Rather than asking a single model to simultaneously extract user preferences, reason about spatial transitions, and produce a final ranking, a multi-agent system assigns each concern to a dedicated agent with a focused prompt and a well-defined output contract. The resulting pipeline is more modular, more interpretable, and more amenable to targeted improvement than a monolithic model.

For next-POI recommendation, the multi-agent paradigm is a natural fit. The prediction task is compositional: it requires understanding who the user is, what has changed

recently, what the local spatial context suggests, and how to integrate these signals into a coherent ranking decision. Existing multi-agent systems for this problem demonstrate that distributing these responsibilities across specialised agents improves over single-agent approaches. However, they remain stateless: no information from one completed prediction carries forward to the next, and every prediction is generated entirely from scratch.

## 1.4 Cognition-Inspired Spatial Reasoning

Human navigation is not stateless. Spatial decisions draw on several distinct types of knowledge that operate at different timescales and serve different functions.

The first is long-term knowledge of personal preferences and habits: which neighbourhoods a person frequents, what categories of venues they favour, and how their daily schedule is typically organised. This knowledge is largely stable across weeks and months.

The second is short-term episodic awareness: the ability to incorporate recent observations such as current location, unfamiliar surroundings, and deviations from established trajectories. This form of knowledge updates continuously with new experiences and becomes particularly important when behaviour diverges from established patterns.

The third is a form of self-monitoring: the ability to recognise when a familiar strategy is not working and to revise it. If a person takes a wrong turn twice using the same heuristic, they will try a different approach on the third attempt. This metacognitive faculty is not about tracking what happened but about evaluating how well one's own reasoning is performing.

These three types of knowledge operate together in real navigation, yet no existing next-POI recommendation system jointly models all three. Long-term preferences are partially captured through training on historical data, but short-term episodic awareness and metacognitive error correction require access to test-time observations. Fixed-parameter models have no such access. This gap between human spatial cognition and current system capabilities is the central motivation for this thesis.

## 1.5 Origin of Work

The starting point for this work was a straightforward empirical observation: strong next-POI systems tend to achieve competitive Hit Rate scores but substantially lower Mean Reciprocal Rank values. The gap indicates that the correct destination is typically present somewhere in the top-10 ranked list but is not being placed at the top position. The problem is not one of retrieval; it is one of ranking quality.

Examining where the ranking errors came from pointed to a deeper pattern. For the same user, errors were directional rather than random. The system would consistently favour one category of venue over another, or consistently under-weight a type of temporal cue, producing the same kind of mistake repeatedly. This is precisely the kind of systematic error that a person would notice and adjust for after the first or second occurrence. The system had no such capacity; it applied the same learned function to every query without any awareness of whether its recent predictions had been accurate.

This raised a design question: could a recommendation system carry information forward across successive predictions, specifically about what the user has been doing during the current session and whether the system's own recent predictions were correct? The difficulty was that existing approaches to adaptive modelling require gradient updates, which means maintaining a training pipeline at inference time, which is impractical in real-time deployment.

The key insight came from recognising that large language models, when given structured natural-language context, can reason from that context without any parameter changes. A cognitive state maintained as natural language inside a prompt is updated by rewriting the prompt, not by running backpropagation. This made it possible to implement both short-term episodic awareness and metacognitive error correction through structured prompt engineering within a multi-agent loop, entirely at inference time and with no modification to model weights.

CogMaPOI is the realisation of these decisions. The experiments in this thesis test whether this architecture, built on top of a consumer-grade instruction-tuned LLM, can consistently outperform both parametric deep learning baselines and the strongest existing multi-agent systems across diverse benchmark datasets.

## 1.6 Problem Statement

The standard deployment pipeline for next-POI recommendation has no channel through which test-session observations can influence subsequent predictions. What the model learned during training is all it will ever apply, regardless of how much the user's behaviour diverges from the training history during evaluation. A better formulation would treat the test session itself as a source of evidence, accumulate that evidence across prediction steps, and use the system's own past prediction errors to guide future decisions.

The problem addressed in this thesis is stated as follows:

> *To develop a cognition-inspired multi-agent framework that reformulates next-POI recommendation as a continual episodic inference process, enabling parameter-free test-time adaptation through identity priors, episodic memory consolidation, and metacognitive reflection.*

The cognitive state must encode three qualitatively distinct types of information: stable long-term user preferences derived from historical data, short-term episodic evidence of what the user has done during the current session, and a record of the system's own prediction quality that generates corrective guidance for subsequent episodes.

## 1.7 Objectives

The specific objectives of this thesis are:

1. To formalise next-POI recommendation as continual episodic inference, where a structured cognitive state evolves across prediction episodes without any parameter updates to the underlying model.
2. To design a five-agent LLM architecture that operationalises long-term preference modelling, short-term episodic adaptation, and metacognitive error correction as distinct, interacting state components in a strictly causal inference loop.
3. To rigorously evaluate the framework against thirteen baselines spanning transformer-based, graph-based, generative, LLM-based, and multi-agent methods across three Foursquare benchmark datasets, measuring performance on both coverage metrics and the ranking-sensitive metric MRR.

4. To isolate the individual contributions of episodic memory and metacognitive reflection to prediction quality through controlled ablation, and to determine whether their joint effect exhibits interaction effects that neither component produces independently.

5. To assess the robustness of the framework across LLM backbones ranging from 3B to 8B parameters, and to determine whether the performance advantage is structural in origin rather than contingent on model scale.

## 1.8 Proposed Approach

CogMaPOI addresses the problem through a five-agent pipeline that maintains three separately managed state components across a three-episode test window.

Before inference begins, a Profiler Agent constructs a structured identity prior from the user's historical trajectory, encoding stable preferences, routine patterns, and a synthesised reasoning hint. A Retrieval Agent concurrently assembles a user-specific candidate set by combining population-level transition statistics with individual visit frequency, weighted adaptively according to the user's exploration profile.

During inference, the system processes the test trajectory in three episodes of ten check-ins each. After each episode, a Memory Agent updates a running natural-language summary of the user's test-session behaviour, capturing category shifts, spatial changes, and deviations from established routines. A Reflection Agent then evaluates the system's prediction for that episode: a correct prediction reinforces the reasoning strategy used; an incorrect prediction triggers a diagnosis of the failure mode and produces a corrective directive for the next episode. A Reranker Agent produces the subsequent prediction by conditioning on the current episode, the identity prior, the episodic memory, and the reflection record accumulated from all prior episodes.

No agent fine-tunes any model at inference time. All adaptation is carried through natural-language state updates encoded in structured prompts. The system evaluated at Episode 3 has completed two full cycles of episodic consolidation and metacognitive reflection before producing its evaluated output.

# Chapter 2

# Literature Review

This chapter surveys the body of work that motivates and contextualises CogMaPOI. The review is organised into four areas: deep learning methods for sequential next-POI recommendation, LLM-based approaches, multi-agent recommendation frameworks, and cognition-inspired continual learning. Each area is examined not only for its contributions but for the specific limitation that drove the field toward the next paradigm. A final section identifies the specific research gaps that this work addresses.

## 2.1 Deep Learning for Next-POI Recommendation

The transition from collaborative filtering to deep sequential models marked a substantial shift in how user mobility is modelled. Early recurrent approaches such as LSTM-based trajectory encoders demonstrated that temporal ordering in check-in sequences carries predictive signal beyond simple co-occurrence statistics. These gave way to self-attention architectures once transformers proved effective on sequential data.

**Transformer-based methods.** SASRec [1] applies unidirectional self-attention over item sequences, allowing the model to selectively attend to the most relevant prior check-ins when predicting the next. BERT4Rec [2] extends this to bidirectional attention, treating the prediction task as a masked item recovery problem and enabling richer context integration. TiCoSeRec [3] augments the transformer backbone with time-interval-aware data augmentation, explicitly encoding temporal gaps between check-ins as features rather than treating the sequence as uniformly spaced. DuoRec [4] addresses representation degeneration in sequential models through contrastive learning,

constructing augmented views of the same trajectory and using them to regularise the embedding space.

These methods collectively establish that sequential modelling over check-in histories is an effective basis for next-POI prediction. However, they all encode preferences statically: the model trained on a user's historical trajectory cannot update its representation when the user's behaviour shifts during the evaluation window.

**Graph-based methods.** GETNext [5] augments transformer-based trajectory encoding with a global trajectory flow map, capturing population-level mobility patterns as a graph structure that conditions individual predictions. CrossDR [6] introduces spatial-temporal disentanglement, separating the spatial and temporal components of mobility into distinct representation spaces to improve cross-context generalisation. MAERec [7] applies graph masked autoencoding to trajectory data, learning high-order dependencies between POIs through a reconstruction objective. These approaches bring richer structural inductive biases to the problem, but share the same static deployment constraint.

**Generative and continuous-time methods.** MTNet [8] models user preferences through a mobility tree structure that organises visited venues by time slot, enabling fine-grained time-of-day reasoning. DiffuRec [9] applies diffusion-based generative modelling to sequential recommendation, framing the prediction task as iterative denoising over the candidate space. POIGDE [10] uses graph differential equations to model continuous-time preference dynamics, allowing preference evolution to be tracked at arbitrary temporal granularity. While these methods extend the expressiveness of the modelling toolkit, they remain single-pass: a prediction is produced once from the available history and cannot be revised based on what the system observes during evaluation.

Across all three sub-families, venues are represented as numeric identifiers. No model in this paradigm can reason about what a venue is, how it relates semantically to other venue types, or what role it serves in everyday routines. This representational ceiling, rather than any lack of architectural sophistication, is what ultimately constrained deep sequential models. Addressing it required a foundation in which broad semantic knowledge about places is available at inference time without being re-estimated from check-in data alone, and that is precisely the capability large language models provided.

## 2.2 LLM-Based POI Recommendation

The emergence of large language models (LLMs) opened a new avenue for next-POI recommendation by enabling natural-language reasoning over mobility data. Rather than representing trajectories as sequences of item IDs, LLM-based methods convert check-in histories into natural-language descriptions and exploit the semantic knowledge and reasoning capabilities encoded in pre-trained models.

LLM4POI [11] was among the first to demonstrate this paradigm at scale. It reformulates user trajectories as structured natural-language prompts and applies zero-shot LLM inference directly to the ranking task, without any task-specific fine-tuning. The results showed that LLMs can recover meaningful predictive signal from trajectory descriptions, particularly for capturing categorical and temporal preferences that are difficult to express as numeric features. LLMmove [12] extended this direction by investigating zero-shot generalisation of LLMs across geographic regions, treating next-POI prediction as a structured generation task.

Subsequent work has refined the prompt engineering side of this approach. GNPR-SID [13] constructs semantic location identifiers that encode venue characteristics into compact descriptions compatible with LLM input formats. QA-POI [14] reframes spatial recommendation as a question-answering task, exploiting the dialogue-style capabilities of instruction-tuned models. Broader surveys [15, 16] have examined the general synergy between language models and location-based services, finding that LLMs are particularly effective at capturing qualitative preference signals that numeric representations miss.

The key limitation of all these methods is statefulness, or the absence of it. Every prediction call generates a fresh prompt from scratch. There is no accumulation of experience across interactions and no mechanism for reflecting on past outputs. A system that gave a wrong recommendation at one point in the test trajectory has no way to incorporate that failure into subsequent predictions.

Beyond statelessness, the single-model structure carries a further limitation. Producing a reliable recommendation requires several qualitatively distinct reasoning steps: constructing a stable user profile from history, analysing the current spatial and temporal context, and discriminating among candidate venues. Asking one model to handle all of these simultaneously within a single prompt means each concern competes for attention, and none receives the focused treatment it requires. Multi-agent

frameworks emerged as a response to this: by assigning each sub-task to a dedicated agent with a well-scoped prompt and a defined output contract, the pipeline reasons about each dimension of the problem with the depth it warrants.

## 2.3 Multi-Agent Recommendation Frameworks

Multi-agent frameworks distribute the recommendation task across specialised components, allowing different aspects of the problem to be handled by purpose-built agents. This decomposition has shown consistent advantages over monolithic single-model approaches on tasks with multiple interacting sub-problems [17, 18].

MACRec [19] introduced role-based agent hierarchies for general recommendation systems, where coordinator and executor agents collaborate through structured message passing. The framework demonstrated that decomposing the recommendation pipeline into distinct reasoning steps, each handled by a specialised agent, produces more consistent and interpretable outputs than end-to-end approaches.

CoMaPOI [20] applied this principle specifically to next-POI prediction. It bridges structured trajectory data with natural-language reasoning by distributing the task across agents responsible for different aspects of the prediction: trajectory encoding, contextual analysis, and final ranking. The framework achieves strong performance on the Foursquare benchmarks and established the multi-agent decomposition as a viable paradigm for this problem.

LMAPOI [21] builds on CoMaPOI's decomposition strategy with an expanded agent architecture, incorporating additional specialisation for temporal and spatial reasoning. Both frameworks represent the current state of the art among multi-agent methods for next-POI recommendation.

However, CoMaPOI and LMAPOI share the same fundamental limitation as all prior methods: there is no unified cognitive state that persists across predictions. Each prediction is produced independently, with no record of what the user has done during the current session and no memory of whether prior predictions were correct. The multi-agent architecture solves the problem of intra-prediction reasoning decomposition but does not address how information should flow across successive predictions.

This limitation points toward a different body of literature. The question of how a system should carry knowledge forward across successive interactions, updating what it

knows about the user and correcting what it knows about its own reasoning, has been studied extensively in continual learning. The mechanisms developed there, and the extent to which they can be applied without the gradient updates they traditionally require, motivate the final area of review.

## 2.4 Cognition-Inspired Continual Learning

Continual learning addresses the challenge of acquiring new knowledge without catastrophically forgetting what was previously learned. The connection to cognitive science is deliberate: human memory systems selectively consolidate new experiences while preserving stable long-term representations, and this selective updating is what continual learning systems attempt to replicate computationally.

Gradient Episodic Memory (GEM) [22] uses stored episodic examples as optimisation constraints, ensuring that gradient updates for new tasks do not increase loss on previously stored examples. This mechanism provides a formal backward-compatibility guarantee between new and old knowledge. Error Sensitivity Modulation (ESMER) [23] refines this by mimicking the brain's tendency to decrease its sensitivity to errors as their magnitude increases. Unlike standard training objectives, ESMER prioritises learning from small but consistent errors rather than large sudden ones, which helps prevent abrupt representation drift when a model encounters task boundaries.

From a cognitive science perspective, prediction error plays a central role in determining what gets consolidated into memory. Larger errors drive stronger encoding through pattern separation mechanisms [24], and unexpected outcomes tend to produce more durable memory traces than outcomes that were anticipated [25]. Schillaci et al. [26] formalise this relationship in a continual learning context: the magnitude of prediction error governs which experiences are prioritised for consolidation and which internal representations need to be reshaped to support knowledge transfer.

These findings motivate a principled approach to test-time adaptation in recommendation. A system that tracks its own prediction errors across successive predictions and uses them to revise its reasoning strategy is implementing a form of metacognitive error monitoring closely analogous to what cognitive science describes. No prior next-POI recommendation system has operationalised this idea. CogMaPOI does so through the Reflection Agent and its associated state component.

## 2.5 Research Gaps

Four distinct gaps emerge from the literature surveyed above.

**Static inference paradigm.** All reviewed methods generate predictions from representations fixed before evaluation begins. The same learned function is applied to every query regardless of what the user does during the test session, providing no channel through which new behavioural evidence can influence subsequent predictions.

**Lack of explicit episodic memory.** No existing approach tracks what a user has actually done since evaluation started. Every prediction draws on the same historical snapshot, so short-term shifts in behaviour, whether a change in category preference, a visit to an unfamiliar area, or a departure from established routine, remain invisible to the system throughout the session.

**Absence of prediction error correction.** No prior system examines its own prediction outcomes or adjusts its reasoning in response to them. When the same ranking error recurs across successive predictions for the same underlying reason, nothing in the system detects the pattern or acts to correct it.

**No parameter-free test-time adaptation.** Continual learning addresses session-level adaptation and error-driven correction, but both require gradient updates during inference, which is impractical in real-time deployment. No prior work achieves either capability through natural-language state updates alone, without modifying model parameters after training.

CogMaPOI is designed to close all four gaps simultaneously through a unified five-agent architecture with a structured cognitive state.

# Chapter 3

# Methodology

This chapter develops the complete technical foundation of CogMaPOI in three parts. The first part establishes the notation used throughout the thesis. The second part formalises the problem as continual episodic inference and defines the cognitive state that the system maintains across prediction steps. The third part describes the five agents that realise this state and the causal inference loop that governs their interaction.

## 3.1 Preliminaries

### 3.1.1 Entities and Notation

Let $\mathcal{U} = \{u_1, \ldots, u_N\}$ be a set of $N$ users and $\mathcal{P} = \{p_1, \ldots, p_M\}$ a set of $M$ points of interest. Each POI $p \in \mathcal{P}$ is described by a four-tuple:

$$p = \langle \mathrm{id}_p,\ \mathrm{lat}_p,\ \mathrm{lon}_p,\ \mathrm{cat}_p \rangle, \tag{3.1}$$

where $\mathrm{id}_p$ is a unique identifier, $(\mathrm{lat}_p, \mathrm{lon}_p)$ are GPS coordinates in decimal degrees, and $\mathrm{cat}_p$ is a semantic category from a fixed taxonomy (e.g., *Food*, *Travel & Transport*, *Nightlife Spot*). A *check-in* records a single visit:

$$x = \langle u,\ p,\ t \rangle, \tag{3.2}$$

where $u \in \mathcal{U}$, $p \in \mathcal{P}$, and $t$ is a Unix timestamp.

For user $u$, all check-ins sorted chronologically before the test period form the *historical trajectory*:

$$\mathcal{T}_u^H = \{(p_{u,1}, t_{u,1}),\ \ldots,\ (p_{u,n}, t_{u,n})\}, \quad t_{u,i} < t_{u,i+1}. \tag{3.3}$$

The *test trajectory* consists of the final 30 check-ins, following the evaluation protocol of CoMaPOI [20]:

$$\mathcal{T}_u^{test} = \{(p_{u,n+1}, t_{u,n+1}),\ \ldots,\ (p_{u,n+30}, t_{u,n+30})\}. \tag{3.4}$$

The temporal boundary between $\mathcal{T}_u^H$ and $\mathcal{T}_u^{test}$ is strict; no test-period information enters the historical context.

### 3.1.2 Episode Structure

The test trajectory is divided into three non-overlapping, chronologically ordered episodes of ten check-ins each:

$$\mathcal{T}_u^{test} = \bigcup_{e=1}^{3} \mathcal{E}_u^{(e)}, \qquad |\mathcal{E}_u^{(e)}| = 10. \tag{3.5}$$

$\mathcal{E}_u^{(1)}$ contains the earliest ten check-ins, $\mathcal{E}_u^{(2)}$ the middle ten, and $\mathcal{E}_u^{(3)}$ the final ten. At each episode $e$, the system makes a prediction, then updates its internal state before proceeding to episode $e{+}1$. The predicted POI is written $\hat{p}_u^{(e)}$; the ground truth is $y_u^{(e)}$, defined as the last check-in in $\mathcal{E}_u^{(e)}$.

### 3.1.3 Agent Prompting Function Notation

All five agents in CogMaPOI produce outputs by constructing structured natural-language prompts and passing them to a shared large language model. The generic notation $\mathcal{F}(\cdot)$ denotes such a prompting call. Agent-specific instantiations are distinguished by descriptive subscripts, summarised in Table 3.1.

All five functions share the same underlying model. Differences between agents are expressed entirely through prompt structure and output schema, not through separate weights or fine-tuning.

TABLE 3.1: Agent-level prompting function notation.

| Symbol | Agent | Role |
|---|---|---|
| $\mathcal{F}_{\text{profile}}$ | Profiler | Builds the identity prior from $\mathcal{T}_u^H$ |
| $\mathcal{F}_{\text{wt}}$ | Retrieval | Derives adaptive fusion weights from $\mathbf{P}_u$ |
| $\mathcal{F}_{\text{mem}}$ | Memory | Updates episodic memory after each episode |
| $\mathcal{F}_{\text{ref}}$ | Reflection | Evaluates predictions and produces corrective signals |
| $\mathcal{F}_{\text{rank}}$ | Reranker | Produces the final POI prediction per episode |

## 3.2 Problem Formulation

The central departure from prior work is the reframing of next-POI prediction from stateless, single-pass inference into a *continual episodic inference* problem grounded in three cognitive abstractions: identity (stable long-term preferences), episodic memory (recent behavioural patterns), and metacognitive reflection (evaluation of past reasoning, reinforcing effective strategies and revising those that failed).

### 3.2.1 Cognitive State Representation

At each episode $e \in \{1, 2, 3\}$, the system maintains a structured cognitive state for user $u$:

$$\mathcal{S}_u^{(e)} = \left(\mathbf{P}_u,\ \mathcal{M}_u^{(e)},\ \mathcal{R}_u^{(e)}\right). \tag{3.6}$$

The three components encode qualitatively distinct types of knowledge, each operating on a different timescale.

$\mathbf{P}_u$ is the *identity prior*: a structured representation of the user's stable, long-term mobility preferences, constructed once from $\mathcal{T}_u^H$ and held fixed throughout inference. It anchors predictions to established habits and prevents drift toward generic, population-level behaviour as the session progresses.

$\mathcal{M}_u^{(e)}$ is the *episodic memory*: a running natural-language summary of the user's observed activity during the test session, initialised to $\mathcal{M}_u^{(0)} = \varnothing$. It captures short-term behavioural shifts that may diverge from the historical baseline encoded in $\mathbf{P}_u$. Updates are unconditional on prediction correctness, since the record of where the user actually went is informative regardless of whether the preceding prediction succeeded.

$\mathcal{R}_u^{(e)}$ is the *metacognitive reflection record*: the system's assessment of its own prediction quality across completed episodes, initialised to $\mathcal{R}_u^{(0)} = \varnothing$. Unlike episodic memory,

which tracks what the user did, the reflection record tracks how the system reasoned. Correct predictions reinforce the effective strategy; incorrect predictions produce a targeted diagnosis and a corrective directive for subsequent episodes. This component has no counterpart in any prior next-POI recommendation system.

### 3.2.2 Per-Episode Prediction Task

Given a candidate set $\mathcal{C}_u \subseteq \mathcal{P}$ assembled before inference begins, the prediction at episode $e$ is:

$$\hat{p}_u^{(e)} = \mathcal{F}_{\text{rank}}\Big(\mathcal{E}_u^{(e)},\ \mathcal{S}_u^{(e-1)},\ \mathcal{C}_u\Big)\,. \tag{3.7}$$

Conditioning on $\mathcal{S}_u^{(e-1)}$ enforces strict temporal causality: episode-$e$ observations cannot influence the episode-$e$ prediction and are incorporated only through the state update that follows. The candidate set reduces the full search space to a tractable subset; its construction is described in Section 3.3.2.

### 3.2.3 State Transition

Following each prediction, the cognitive state transitions:

$$\mathcal{S}_u^{(e-1)} \longrightarrow \mathcal{S}_u^{(e)}. \tag{3.8}$$

The identity prior is fixed throughout, preserving long-term preference information against short-term test-session noise. The episodic memory and reflection record are updated by their respective agents using the newly observed check-ins and the prediction outcome. All state evolution occurs through natural-language transformations encoded in LLM prompts, with no gradient computation or parameter modification at any stage.

## 3.3 Proposed Methodology

CogMaPOI comprises five LLM-based agents arranged in a strictly causal loop. The Profiler and Retrieval agents execute once per user before inference begins; the Memory, Reflection, and Reranker agents each run once per episode. Figure 3.1 shows the complete pipeline.

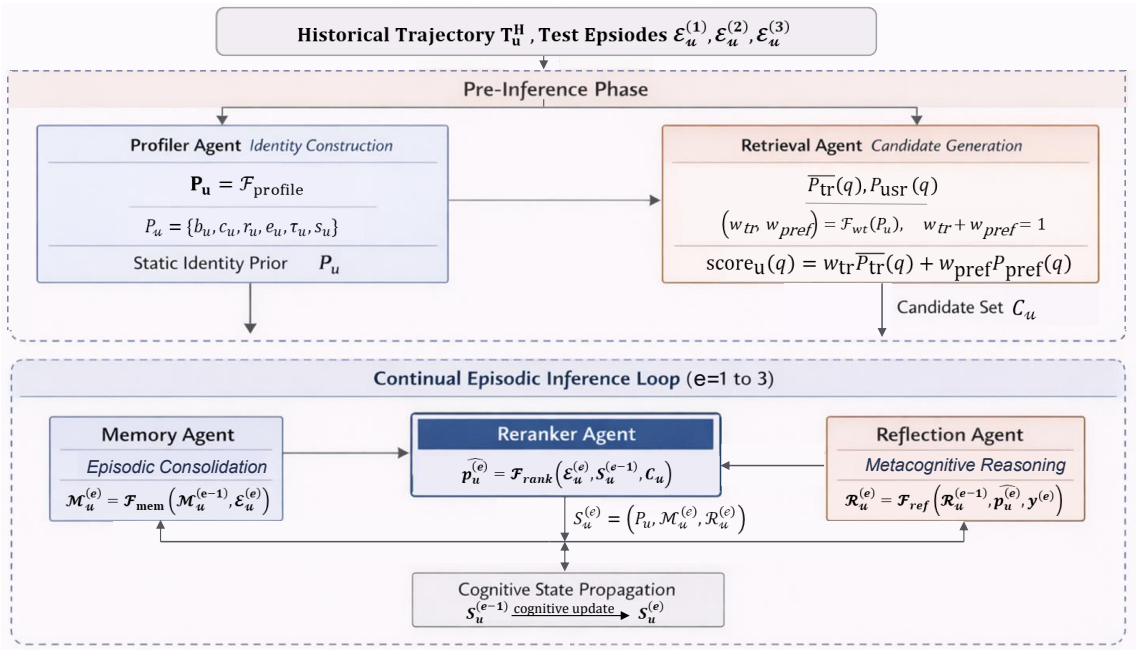


FIGURE 3.1: CogMaPOI architecture. The pre-inference phase (top) runs once per user. The episodic loop (bottom) executes three times, with Memory, Reranker, and Reflection agents updating the cognitive state at each iteration.

The design is motivated by a specific observation about human spatial cognition: people navigate using stable long-term habits, awareness of where they have been recently, and the capacity to recognise and correct their own errors. CogMaPOI assigns one dedicated agent to each of these three faculties, using separate state components so that updates to one do not interfere with the logic of another.

All agents share a single instruction-tuned LLM with no agent-specific fine-tuning. Performance differences attributable to the framework (Chapter 5) therefore reflect the architectural design and the cognitive state structure, not any parametric advantage from task-specific training. The choice of a single shared model also keeps the system deployable on consumer-grade hardware, as demonstrated by the experimental setup in Chapter 4.

### 3.3.1 Profiler Agent: Identity Construction

The Profiler runs once per user before any inference begins. It takes the full historical trajectory $\mathcal{T}_u^H$ as input and produces the identity prior $\mathbf{P}_u$ as a structured JSON object:

$$\mathbf{P}_u = \mathcal{F}_{\text{profile}}(\mathcal{T}_u^H) = \{b_u,\ c_u,\ r_u,\ e_u,\ \tau_u,\ s_u\}. \tag{3.9}$$

The six fields collectively characterise the user at a level of abstraction that is both human-readable and directly usable by subsequent agents.

**Behavioural summary ($b_u$).** A free-text description of the user's mobility personality, capturing qualitative tendencies such as a preference for consistent routines versus spontaneous exploration, or a concentration of visits in a small geographic radius versus wide spatial dispersion.

**Preferred categories ($c_u$).** A frequency-ranked list of the POI semantic categories most visited by the user, providing a compact categorical profile that guides the Reranker when discriminating between candidates of different venue types.

**Routine patterns ($r_u$).** Daily and weekly visitation cycles encoded as structured dictionaries. A user who consistently visits coffee shops on weekday mornings and transit hubs on evenings exhibits a strongly structured routine that predicts large portions of their future behaviour.

**Exploration style ($e_u$).** A ternary value $e_u \in \{\text{Low, Moderate, High}\}$ computed from the ratio of unique POIs visited to total check-ins. This field plays a direct quantitative role in retrieval weighting, as described in the following subsection.

**Temporal habits ($\tau_u$).** A mapping from time-of-day and day-of-week slots to characteristic venue categories, enabling time-sensitive predictions when the episode timestamps suggest a particular temporal context.

**Strategy hint ($s_u$).** A meta-level reasoning directive synthesised from all five preceding fields, such as "prioritise transit and food venues on weekday mornings; deprioritise entertainment venues during work hours." This hint is injected directly into the Reranker's prompt to reduce the inferential load required at prediction time.

Natural-language representations are used throughout rather than numeric vectors so that all profile fields can be inserted directly into LLM prompts without any embedding or transformation step. This design choice is what makes the system applicable on top of any instruction-tuned LLM without retraining.

### 3.3.2 Retrieval Agent: Candidate Construction

Scoring all $M$ POIs through an LLM at every prediction step is not feasible ($M$ ranges from 5,086 to 13,564 across datasets). A user-specific candidate set $\mathcal{C}_u \subseteq \mathcal{P}$ is assembled from two efficient statistical signals before inference begins.

**Transition model.** First-order Markov transition probabilities over consecutive POI pairs in the training corpus:

$$P_{\text{tr}}(q \mid p) = \frac{N(p,q)}{\sum_j N(p,j)}, \tag{3.10}$$

where $N(p,q)$ counts $p \to q$ transitions across all users. Scores from the $L$ most recent check-ins are averaged to smooth the directional signal without losing recency:

$$\bar{P}_{\text{tr}}(q) = \frac{1}{L}\sum_{i=0}^{L-1} P_{\text{tr}}(q \mid p_{n-i}), \qquad L = 2. \tag{3.11}$$

**User preference model.** Normalised historical visit frequency for user $u$:

$$P_{\text{pref}}(q) = \frac{N_u(q)}{|\mathcal{T}_u^H|}, \tag{3.12}$$

which retains locations with personal significance to the user even when population-level transition patterns do not suggest them.

**Profile-conditioned fusion.** The appropriate balance between the two signals depends on the user's exploration style. Low-exploration users revisit familiar locations frequently, so the preference model is more predictive for them. High-exploration users frequently visit new locations, so the transition model is more useful. An LLM call on $\mathbf{P}_u$ produces adaptive fusion weights:

$$(w_{\text{tr}},\ w_{\text{pref}}) = \mathcal{F}_{\text{wt}}(\mathbf{P}_u), \qquad w_{\text{tr}} + w_{\text{pref}} = 1. \tag{3.13}$$

The composite score and resulting candidate set are:

$$\text{score}_u(q) = w_{\text{tr}}\bar{P}_{\text{tr}}(q) + w_{\text{pref}}P_{\text{pref}}(q), \tag{3.14}$$

$$\mathcal{C}_u = \text{Top-}K(\text{score}_u(q))\,, \qquad K = 125. \tag{3.15}$$

$\mathcal{C}_u$ is fixed before inference and does not change across episodes. The selection of $K = 125$ and $L = 2$ is justified empirically in Section 5.4.

### 3.3.3 Memory Agent: Episodic Consolidation

The Memory Agent maintains the episodic memory $\mathcal{M}_u^{(e)}$, initialised to $\mathcal{M}_u^{(0)} = \varnothing$ before Episode 1. After each episode, the agent incorporates the ten newly observed check-ins from $\mathcal{E}_u^{(e)}$ into the existing summary:

$$\mathcal{M}_u^{(e)} = \mathcal{F}_{\text{mem}}\left(\mathcal{M}_u^{(e-1)},\ \mathcal{E}_u^{(e)}\right). \tag{3.16}$$

The update is a natural-language paragraph that integrates existing knowledge of the user's test-session behaviour with patterns from the most recent ten check-ins. It notes changes in visited categories relative to historical baselines, new spatial areas not present in the historical trajectory, and deviations from the routines in the identity prior. Output length is bounded to prevent unbounded accumulation.

A key design decision is that this update is unconditional on prediction correctness. The Memory Agent receives only the actual observed check-ins, not the system's prediction; its job is to track the user, not to evaluate the system. This separation ensures that behavioural evidence is always incorporated regardless of whether the preceding prediction succeeded or failed.

### 3.3.4 Reflection Agent: Metacognitive Reasoning

After each episode, the Reflection Agent compares the system's prediction against the ground truth and updates the reflection record from $\mathcal{R}_u^{(0)} = \varnothing$:

$$\mathcal{R}_u^{(e)} = \mathcal{F}_{\text{ref}}\left(\mathcal{R}_u^{(e-1)},\ \hat{p}_u^{(e)},\ y_u^{(e)}\right). \tag{3.17}$$

A correct prediction reinforces the reasoning strategy that produced it, accumulating a record of which approaches, such as prioritising temporal context or trusting transition signals over frequency counts, have proven reliable for this user.

An incorrect prediction triggers a diagnostic. The agent examines the relationship between the predicted POI and the ground truth, considering factors such as category mismatch, geographic proximity, and temporal context, and identifies the specific failure mode. The diagnosis is issued as an imperative directive, for example "the model ranked candidates from the Office category above candidates from the Food category, but this user visits food venues during the late afternoon on weekdays." This directive is injected into the

subsequent Reranker prompt, directly adjusting which signals the Reranker weights more heavily.

The distinction between memory and reflection is not redundant. Memory enriches the Reranker's contextual understanding of the user. Reflection changes how the Reranker reasons, by providing direct guidance about which strategies to apply and which to avoid. The ablation study in Section 5.2 confirms that removing each component produces a categorically different pattern of degradation.

### 3.3.5 Reranker Agent: Inference and Integration

At each episode $e$, the Reranker integrates four inputs to produce the prediction:

$$\hat{p}_u^{(e)} = \mathcal{F}_{\text{rank}}\Big(\mathcal{E}_u^{(e)},\ \mathcal{S}_u^{(e-1)},\ \mathcal{C}_u\Big)\,. \tag{3.18}$$

The current episode $\mathcal{E}_u^{(e)}$ provides the most immediately relevant context: what the user has done in the last ten visits, including categories, locations, and timestamps. The prior cognitive state $\mathcal{S}_u^{(e-1)} = (\mathbf{P}_u, \mathcal{M}_u^{(e-1)}, \mathcal{R}_u^{(e-1)})$ brings in the stable identity prior, the episodic record of test-session behaviour, and the metacognitive assessment of reasoning quality. The conditioning on $e-1$ rather than $e$ enforces strict temporal causality: no episode-$e$ information can influence the episode-$e$ prediction. The candidate set $\mathcal{C}_u$ focuses the model's reasoning on discrimination among plausible options rather than recall from the full vocabulary.

The Reranker prompt instructs the model to rank candidates by accounting for temporal context, categorical preferences, spatial patterns, and the corrective directives from the reflection record, using the strategy hint $s_u$ as a synthesised starting point. The top-ranked candidate is taken as $\hat{p}_u^{(e)}$.

### 3.3.6 The Continual Episodic Inference Loop

Algorithm 1 formalises the complete inference procedure.

Four properties characterise the loop.

**Algorithm 1** Continual Inference in CogMaPOI

**Require:** $\mathcal{E}_u^{(1)}, \mathcal{E}_u^{(2)}, \mathcal{E}_u^{(3)}$; $\mathbf{P}_u$; $\mathcal{C}_u$

$\mathcal{M}_u^{(0)} \leftarrow \varnothing$; $\mathcal{R}_u^{(0)} \leftarrow \varnothing$; $\mathcal{S}_u^{(0)} \leftarrow (\mathbf{P}_u, \mathcal{M}_u^{(0)}, \mathcal{R}_u^{(0)})$
**for** $e = 1, 2, 3$ **do**
  $\hat{p}_u^{(e)} \leftarrow \mathcal{F}_{\text{rank}}(\mathcal{E}_u^{(e)}, \mathcal{S}_u^{(e-1)}, \mathcal{C}_u)$
  $\mathcal{M}_u^{(e)} \leftarrow \mathcal{F}_{\text{mem}}(\mathcal{M}_u^{(e-1)}, \mathcal{E}_u^{(e)})$
  $\mathcal{R}_u^{(e)} \leftarrow \mathcal{F}_{\text{ref}}(\mathcal{R}_u^{(e-1)}, \hat{p}_u^{(e)}, y_u^{(e)})$
  $\mathcal{S}_u^{(e)} \leftarrow (\mathbf{P}_u, \mathcal{M}_u^{(e)}, \mathcal{R}_u^{(e)})$
**end for**
**Evaluate** on $\hat{p}_u^{(3)}$

**Temporal causality.** All predictions are conditioned on $\mathcal{S}_u^{(e-1)}$, so episode-$e$ observations cannot affect the episode-$e$ prediction. The ground truth $y_u^{(e)}$ enters the system only through $\mathcal{F}_{\text{ref}}$, strictly after the prediction is committed.

**Progressive enrichment.** By Episode 3, the cognitive state has undergone two full consolidation and reflection cycles. The Reranker at Episode 3 therefore operates with substantially richer contextual and diagnostic information than it had at Episode 1.

**Parameter-free adaptation.** No gradient computation or model fine-tuning occurs at any point. All adaptation is carried through natural-language state updates encoded in structured prompts, making the framework deployable without any training infrastructure.

**Evaluation fairness.** Metric computation is restricted to Episode 3 predictions alone. Episodes 1 and 2 function as cognitive warm-up cycles that build the state used at Episode 3; evaluating them would not yield a fair comparison with single-shot baselines, which process the full 30-check-in window in a single forward pass with no iterative warm-up. In both cases, the same 30 check-ins are observed before committing a prediction, and neither system has access to the ground truth for the episode being evaluated.

# Chapter 4

# Implementation Details

## 4.1 Hardware and Software

All experiments run on a single NVIDIA RTX 2000 Ada GPU (16 GB VRAM) under Ubuntu 22.04, CUDA 12.1, and Python 3.10. The inference stack uses PyTorch 2.1.0, Hugging Face `transformers` 4.40.0 for model loading, `bitsandbytes` 0.43.0 for quantisation, and `accelerate` 0.28.0 for device management.

## 4.2 Model Configuration

The primary backbone is Llama 3.1-8B-Instruct, chosen because the Profiler requires structured JSON output and the Reranker requires a ranked candidate list; both are output format constraints that base models without instruction tuning do not reliably satisfy.

The model is loaded with 4-bit NF4 quantisation (FP16 compute dtype, double quantisation enabled), reducing the 8B-parameter model to approximately 5.5 GB of VRAM and leaving adequate headroom for the KV cache. Generation uses temperature 0.2 and a maximum of 200 output tokens per call. Two inference passes are run for each Reranker call; ranked outputs are averaged across the two passes and the candidate with the lowest average rank is selected. This halves stochastic variance in the top-1 prediction without meaningful overhead. Memory and Reflection agents use a single pass, as phrasing variation in their outputs does not affect downstream prediction quality.

## 4.3 Datasets

Three Foursquare check-in datasets from the CoMaPOI benchmark [2] are used. Each record contains a user ID, POI ID, semantic category, GPS coordinates, and timestamp. Table 4.1 gives summary statistics.

TABLE 4.1: Summary statistics for the three Foursquare datasets.

| Dataset | Users | POIs | Check-ins | Avg. check-ins/user | Period |
|---|---|---|---|---|---|
| NYC | 988 | 5,086 | 99,964 | 101.2 | Apr 2012 – Feb 2013 |
| TKY | 2,206 | 7,849 | 325,313 | 147.5 | Apr 2012 – Feb 2013 |
| CA | 1,818 | 13,564 | 174,791 | 96.1 | Feb 2009 – Oct 2010 |

**NYC** is a dense urban dataset with compact geography. **TKY** exhibits strong trajectory regularity driven by train-based commuting, making it the most predictable of the three for sequential models. **CA** spans a geographically dispersed, multi-city region with the largest POI vocabulary (13,564), creating the hardest retrieval conditions. Consistent performance across all three is a more demanding bar than on any single dataset.

## 4.4 Data Preprocessing and Episode Construction

Following CoMaPOI [20], check-ins for each user are sorted chronologically. The final 30 constitute the test trajectory; all earlier check-ins form the historical trajectory. Users with fewer than 30 total check-ins are excluded. The test trajectory is split into three ten-check-in episodes per Section 3.1.2; the ground truth for each episode is the tenth check-in. Candidate set construction uses only the historical trajectory: transition counts $N(p, q)$ are computed globally across all users, while $P_{\text{pref}}(q)$ is per-user.

## 4.5 Evaluation Metrics

Three standard retrieval metrics are computed on Episode 3 predictions only (see Section 3.3.6), at cut-offs $k \in \{5, 10\}$ per the CoMaPOI protocol.

**HR@$k$** (Hit Rate) is the fraction of users for whom the ground truth appears in the top-$k$ list:

$$\text{HR@}k = \frac{1}{|\mathcal{U}_{test}|} \sum_{u} \mathbf{1}\left[y_u^{(3)} \in \text{Top-}k(\hat{\mathbf{r}}_u)\right]. \tag{4.1}$$

**NDCG@$k$** discounts correct answers by rank position:

$$\text{NDCG@}k = \frac{1}{|\mathcal{U}_{test}|} \sum_u \frac{\mathbf{1}\Big[\text{rank}(y_u^{(3)}) \leq k\Big]}{\log_2(\text{rank}(y_u^{(3)}) + 1)}. \tag{4.2}$$

**MRR** (Mean Reciprocal Rank) assigns each user $1/\text{rank}(y_u^{(3)})$:

$$\text{MRR} = \frac{1}{|\mathcal{U}_{test}|} \sum_u \frac{1}{\text{rank}(y_u^{(3)})}. \tag{4.3}$$

HR@$k$ is satisfied by any placement in the top-$k$, so it is insensitive to rank quality within that range. NDCG@$k$ rewards higher-ranked correct answers. MRR is the most discriminative for systems that aim to place the correct POI at rank 1: a prediction at rank 2 contributes 0.5, at rank 3 contributes 0.33. The MRR-versus-HR gap observed in the results (Chapter 5) reflects that CogMaPOI's primary effect is on rank quality rather than raw coverage.

## 4.6 Baselines and Comparison Protocol

CogMaPOI is compared against thirteen baselines across five categories: transformer-based (SASRec [1], BERT4Rec [2], TiCoSeRec [3]), graph-based (GETNext [5], CrossDR [6], MAERec [7], DuoRec [4]), generative (POIGDE [10], MTNet [8], DiffuRec [9]), LLM-based (LLM4POI [11]), and multi-agent (CoMaPOI [20], LMAPOI [21]). All baseline figures are taken directly from the CoMaPOI unified evaluation benchmark [20], which uses identical data splits, candidate sets, and evaluation cut-offs. The hyperparameters $K = 125$ and $L = 2$ are fixed from the retrieval analysis on NYC before any comparative evaluation and applied uniformly across all three datasets.

# Chapter 5

# Results and Analysis

This chapter analyses the experimental results across four complementary lenses: comparative performance against thirteen baselines (Section 5.1), controlled ablations isolating the individual contributions of episodic memory and metacognitive reflection (Section 5.2), backbone sensitivity across a range of LLM sizes (Section 5.3), and retrieval hyperparameter analysis (Section 5.4). Section 5.5 synthesises the cross-cutting patterns that emerge.

## 5.1 Comparative Performance Analysis

Table 5.1 reports the full comparative results. Before examining per-dataset specifics, the most analytically significant feature of the table is a consistent asymmetry: MRR improvements are substantially larger than HR improvements across all three datasets. Understanding what drives this asymmetry is central to interpreting what CogMaPOI actually does.

### 5.1.1 The MRR-versus-HR Asymmetry

On NYC, MRR improves 56.88% over the strongest baseline while HR@10 improves only 8.28%. On TKY, MRR improves 46.36% while HR@10 improves 2.00%. On CA, the divergence is greatest, with MRR gaining 89.35% against HR@10's 44.72%. This pattern is not incidental; it traces directly to what the cognitive adaptation mechanisms are designed to accomplish.

TABLE 5.1: Performance comparison across all datasets (%). N@$k$ = NDCG@$k$. $\Delta$ denotes relative improvement over the best baseline for each metric.

| | NYC | | | | | TKY | | | | | CA | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | HR@5 | HR@10 | N@5 | N@10 | MRR | HR@5 | HR@10 | N@5 | N@10 | MRR | HR@5 | HR@10 | N@5 | N@10 | MRR |
| SASRec | 40.38 | 48.28 | 28.98 | 31.26 | 27.01 | 34.72 | 43.25 | 25.48 | 28.37 | 24.70 | 21.34 | 26.40 | 16.04 | 17.68 | 15.94 |
| BERT4Rec | 39.60 | 46.70 | 29.03 | 30.94 | 27.11 | 36.17 | 43.47 | 26.70 | 29.08 | 25.39 | 23.22 | 28.22 | 17.26 | 18.66 | 16.61 |
| DuoRec | 34.72 | 40.69 | 24.54 | 26.34 | 22.76 | 24.66 | 31.19 | 18.77 | 20.89 | 18.67 | 19.47 | 24.04 | 14.47 | 15.85 | 14.31 |
| TiCoSeRec | 40.69 | 48.08 | 29.65 | 32.04 | 27.54 | 29.33 | 37.04 | 21.83 | 24.21 | 21.51 | 22.66 | 28.11 | 16.99 | 18.48 | 16.75 |
| CrossDR | 40.80 | 47.20 | 30.67 | 32.59 | 28.66 | 32.51 | 40.57 | 23.58 | 26.19 | 22.65 | 22.11 | 28.61 | 16.11 | 18.18 | 15.79 |
| MAERec | 37.65 | 44.53 | 27.72 | 29.85 | 26.10 | 32.73 | 40.66 | 23.93 | 26.14 | 23.05 | 24.37 | 29.37 | 17.76 | 19.48 | 17.09 |
| GETNext | 43.60 | 51.30 | 32.56 | 35.13 | 30.97 | 41.11 | 48.50 | 30.75 | 33.21 | 29.21 | 24.26 | 29.32 | 17.83 | 19.48 | 17.16 |
| POIGDE | 34.11 | 38.36 | 26.64 | 27.97 | 25.40 | 28.20 | 31.82 | 22.46 | 23.57 | 21.80 | 22.61 | 26.73 | 18.24 | 19.49 | 17.89 |
| MTNet | 47.57 | 53.24 | 35.85 | 37.69 | 33.40 | 44.15 | 51.22 | 32.99 | 35.37 | 31.18 | 30.25 | 36.08 | 22.39 | 24.27 | 21.35 |
| DiffuRec | 37.25 | 40.79 | 29.16 | 30.23 | 27.74 | 34.50 | 39.89 | 26.89 | 28.70 | 25.94 | 21.67 | 24.15 | 17.35 | 18.06 | 16.60 |
| LLM4POI | 44.64 | 46.66 | 37.49 | 38.16 | 35.90 | 34.90 | 39.48 | 27.23 | 28.22 | 29.72 | 23.65 | 25.91 | 19.01 | 19.39 | 20.13 |
| CoMaPOI | 51.62 | 59.01 | 40.42 | 42.82 | 37.67 | 45.83 | 54.26 | 34.48 | 37.20 | 31.82 | 33.00 | 39.16 | 24.96 | 26.96 | 23.10 |
| LMAPOI | 54.56 | 64.66 | 42.56 | 45.75 | 40.28 | 48.26 | 57.52 | 36.26 | 39.49 | 34.13 | 31.43 | 37.23 | 24.30 | 26.17 | 23.11 |
| **CogMaPOI** | **64.67** | **70.01** | **51.97** | **53.76** | **63.19** | **48.67** | **58.67** | **37.13** | **40.44** | **49.95** | **46.67** | **56.67** | **33.60** | **36.85** | **43.76** |
| $\Delta$ (rel.%) | +18.53 | +8.28 | +22.11 | +17.51 | +56.88 | +0.85 | +2.00 | +2.40 | +2.41 | +46.36 | +41.42 | +44.72 | +34.62 | +36.69 | +89.35 |

HR@$k$ treats any placement within the top-$k$ list as equivalent. A system placing the correct POI at rank 8 receives the same credit as one placing it at rank 1, provided $k \geq 8$. MRR, by assigning reciprocal rank scores, penalises every position below first. The episodic loop does not alter which candidates are retrieved; the candidate set is fixed before inference begins and does not change across episodes. What the loop does alter is the ordering among those candidates. Episodic memory gives the Reranker a richer and more current description of the user's behaviour, helping it weigh candidates more accurately. The reflection record goes further by directly correcting the ranking errors made in prior episodes. Both mechanisms operate entirely within the retrieved set, reshaping how candidates are ordered rather than which candidates are present. MRR is the metric most sensitive to exactly this kind of improvement, which explains why its gains are consistently and substantially larger than HR gains across all three datasets and all five reported metrics.

### 5.1.2 Dataset-Level Analysis

The pattern of improvement across the three datasets is not uniform, and the variation itself reveals something analytically meaningful about when the cognitive adaptation framework is most valuable.

On **NYC**, the gains are broadly distributed. CogMaPOI reaches an MRR of 63.19%, compared to 40.28% for the strongest baseline LMAPOI, a relative gain of 56.88%. The NDCG improvements follow a similar direction: NDCG@5 rises from 42.56% to 51.97% and NDCG@10 from 45.75% to 53.76%, representing relative gains of 22.11% and 17.51% respectively. These NDCG results confirm that the improvement is not concentrated

only at rank 1. The cognitive state shifts the correct POI toward the top of the ranked list in a graded manner, benefiting both rank-1 placement (captured by MRR) and placement within the top-5 (captured by NDCG@5). The prior multi-agent systems achieve competitive HR values because their decomposition strategy already retrieves the correct POI fairly often, but their MRR and NDCG values are substantially lower, pointing to a consistent inability to rank retrieved candidates in the right order. CogMaPOI's reflection record corrects this specific failure.

On **TKY**, the dataset's structural regularity constrains HR and NDCG improvements but leaves ample room for MRR gains. The strong sequential patterns in Tokyo commuter trajectories mean that transition-based retrieval is already effective at placing the correct POI into the candidate set, and high coverage compresses HR and NDCG differences between methods. Yet MRR improves 46.36%, the second largest gain across datasets. This confirms that even in highly regular trajectory settings, static models produce consistent ranking errors that episodic context and reflective correction can address. When the correct POI is routinely present but routinely mis-ranked, the reflection mechanism's targeted corrective guidance matters most.

On **CA**, CogMaPOI achieves its most substantial improvements across all five metrics simultaneously: HR@5 rises by 41.42%, NDCG@5 by 34.62%, NDCG@10 by 36.69%, and MRR by 89.35%. The CA dataset contains 13,564 unique POIs spread across a geographically dispersed region, a vocabulary nearly three times larger than NYC. In settings with this degree of spatial heterogeneity, user individuation becomes the primary source of predictive signal. The identity prior $\mathbf{P}_u$ encodes precisely the kind of user-specific geographic and categorical preferences that distinguish one user from the broader population, and this signal becomes more discriminative as the search space grows. Episodic memory adds further value because behavioural drift matters more when users have a wider range of plausible next destinations. Both the NDCG and MRR gains on CA suggest that the cognitive state helps the system not only identify the correct POI but place it substantially higher in the ranking than any baseline manages.

## 5.2 Ablation Study

Table 5.2 presents the ablation results on NYC. The degradation patterns produced by removing each component are qualitatively different, and that difference is as informative as the magnitude of the drop.

Table 5.2: Ablation study on NYC (%). Relative decrease from full CogMaPOI shown in parentheses.

| Configuration | HR@5 | HR@10 | NDCG@5 | NDCG@10 | MRR |
|---|---|---|---|---|---|
| **CogMaPOI (full)** | 64.67 | 70.01 | 51.97 | 53.76 | 63.19 |
| w/o Memory | 60.02 (−7.19%) | 68.09 (−2.74%) | 47.33 (−8.93%) | 49.84 (−7.29%) | 57.96 (−8.27%) |
| w/o Reflection | 57.01 (−11.85%) | 69.15 (−1.23%) | 48.65 (−6.39%) | 52.98 (−1.45%) | 62.12 (−1.69%) |
| w/o Both | 51.76 (−19.97%) | 63.18 (−9.75%) | 38.81 (−25.32%) | 41.59 (−22.64%) | 50.66 (−19.83%) |

### 5.2.1 Memory as Broad Contextual Enrichment

Removing episodic memory produces a proportional decline distributed broadly across all five metrics. HR@5 falls 7.19%, HR@10 falls 2.74%, NDCG@5 falls 8.93%, NDCG@10 falls 7.29%, and MRR falls 8.27%. The near-uniform spread of these reductions is itself an analytical finding. Where a mechanism that specifically improves top-1 placement would produce disproportionate MRR degradation, and where one that specifically improves top-5 placement would produce disproportionate HR@5 and NDCG@5 degradation, the memory ablation shows no such concentration. This breadth reflects the nature of memory's contribution: it functions as a context enrichment layer that helps the Reranker reason better across the full ranked list rather than targeting any particular rank depth.

Without episodic memory, the Reranker at Episode 3 operates solely on the current ten check-ins and the static identity prior. The identity prior encodes historical preferences that may no longer reflect the user's current situation. A user whose recent session diverges from their historical baseline, visiting a different part of the city or a different category of venue, provides no corrective signal to a system that cannot accumulate what it observes. Memory closes this gap by maintaining a running description of test-session behaviour that updates unconditionally after each episode.

### 5.2.2 Reflection as a Top-Rank Corrector

Removing reflection produces a sharply concentrated pattern. HR@5 falls 11.85%, the largest single-component degradation in the entire study, while HR@10 falls only 1.23% and MRR falls only 1.69%. The NDCG metrics tell the same story from a complementary

angle: NDCG@5 falls 6.39%, indicating that top-5 quality suffers meaningfully, while NDCG@10 falls only 1.45%, confirming that deeper in the ranked list the system is nearly unchanged.

This pattern has a precise interpretation. HR@5 and HR@10 differ solely in whether the ground-truth POI lands within positions 1 through 5 versus positions 1 through 10. NDCG@5 and NDCG@10 differ in how much credit they assign to placements in the lower half of the top-10, with NDCG@5 weighting the very top positions more heavily. A large fall in HR@5 and NDCG@5 with minimal fall in HR@10 and NDCG@10 together imply that removing reflection moves the correct POI from within the top-5 into positions 6 through 10. The candidate is still being retrieved; the Reranker is simply no longer promoting it to the positions that matter most for ranking quality.

This is exactly the failure mode the Reflection Agent addresses. It diagnoses specific ranking errors from prior episodes, such as systematically placing the wrong category at the top or over-weighting venue frequency relative to time-of-day context, and issues targeted directives to the subsequent Reranker. Without this corrective signal, the system defaults to a ranking strategy that is plausible in general but poorly calibrated for the individual user, producing the concentrated degradation in HR@5 and NDCG@5 that the ablation reveals.

### 5.2.3 Synergistic Effect of Memory and Reflection

Removing both components simultaneously produces degradations that exceed the sum of the individual effects for most metrics. NDCG@5 falls 25.32% jointly, compared to 8.93% and 6.39% separately, summing to 15.32%. NDCG@10 falls 22.64% jointly versus a combined total of 8.74% individually. MRR falls 19.83% jointly versus 9.96% cumulatively. The joint HR@5 drop of 19.97% also exceeds the sum of 7.19% and 11.85%.

The reason for this pattern is that the two components are not independent. The quality of the reflection diagnosis depends on what information the Reranker had when it made the error being diagnosed. Richer episodic context means the Reranker was reasoning from better information, which in turn makes it possible to give a more precise account of why the prediction still failed. When both are removed, the system loses not only both forms of adaptation independently, but also the reinforcing relationship between them. The resulting degradation, particularly in the NDCG metrics where both depth and rank position are jointly penalised, is substantially larger than what either removal alone

TABLE 5.3: Backbone sensitivity analysis on NYC (%). All configurations use the same CogMaPOI architecture with $K$=125 and $L$=2.

| Backbone | HR@5 | HR@10 | NDCG@5 | NDCG@10 | MRR |
|---|---|---|---|---|---|
| **Llama 3.1-8B** | **64.67** | **70.01** | **51.97** | **53.76** | **63.19** |
| Mistral 7B | 62.07 | 67.76 | 49.63 | 50.94 | 60.08 |
| Qwen 2.5-5B | 61.26 | 67.18 | 49.18 | 50.35 | 59.35 |
| Qwen 1.5-7B | 60.43 | 66.72 | 48.69 | 49.71 | 58.92 |
| Llama 3.2-3B | 59.85 | 65.54 | 47.65 | 48.22 | 57.81 |
| Qwen 2.5-3B | 58.09 | 62.91 | 45.78 | 46.57 | 55.99 |

predicts. The joint ablation confirms that memory and reflection are coupled components of the adaptive reasoning process rather than independent additions.

## 5.3 LLM Backbone Sensitivity

The backbone results in Table 5.3 demonstrate a consistent gradation: performance decreases as backbone capacity is reduced, but the degradation is gradual rather than precipitous. Qwen 2.5-3B, the smallest model tested, achieves HR@5 of 58.09%, NDCG@5 of 45.78%, and MRR of 55.99%. Each of these substantially exceeds the corresponding scores of LMAPOI, the strongest static baseline, which records HR@5 of 54.56%, NDCG@5 of 42.56%, and MRR of 40.28%. The margin is not marginal: even the weakest CogMaPOI configuration outperforms the best-performing prior multi-agent system by 3–15 percentage points across metrics. This establishes that the performance advantage is structural in origin and not an artefact of the backbone's raw capacity.

Two comparisons within the table are analytically interesting because they isolate the effect of instruction-following alignment from raw parameter count. Qwen 1.5-7B scores HR@5 of 60.43% and NDCG@5 of 48.69%, both lower than Qwen 2.5-5B (HR@5: 61.26%, NDCG@5: 49.18%) despite having more parameters. The Qwen 2.5 training pipeline introduced substantially improved alignment, and this matters for CogMaPOI because every agent relies on the model's ability to follow structured output specifications: the Profiler must produce valid JSON, the Reranker must produce a ranked list, and the Reflection Agent must distinguish between correct and incorrect prediction scenarios and format its diagnosis appropriately. A model that deviates from these formats degrades every downstream component. The gap between Mistral 7B and Llama 3.1-8B tells the same story: a 3.86% difference in HR@5 and a 4.93% difference in MRR between models of

comparable size, attributable to the stronger instruction tuning of Llama 3.1-8B-Instruct rather than to any advantage in raw capacity.

## 5.4 Retrieval Stage Analysis

Both retrieval hyperparameters are selected on the basis of final ranking metrics rather than retrieval recall alone, since high recall with a poorly ordered candidate set can improve HR while simultaneously degrading MRR. The analysis uses NYC.

### 5.4.1 Candidate Set Size

The recall values in Table 5.4 stabilise from $K$=75 onward, meaning that the Markov transition model and user preference model together identify a fixed proportion of ground-truth POIs as high-scoring candidates regardless of how much further the candidate set is expanded.

Within this plateau, the ranking metrics continue to change, and their trajectory is instructive. NDCG@10 and MRR both increase from $K$=75 to $K$=125, indicating that adding candidates in this range introduces genuine but lower-probability correct options without yet introducing enough distractors to disrupt the Reranker's ability to discriminate. Above $K$=125, the trend reverses. At $K$=150, MRR falls to 61.15% and NDCG@10 falls to 51.65%, below the $K$=125 values of 63.19% and 53.76%. At $K$=200, HR@10 nearly recovers to 69.84%, but MRR remains depressed at 62.72%, a pattern that precisely mirrors the MRR-versus-HR asymmetry seen in the comparative results: recovering more ground-truth POIs into the top-10 does not improve ranking quality at the top.

The $K$=25 figures require careful interpretation. MRR at $K$=25 appears competitive at 60.87%, but nearly 36% of users have no viable ground-truth candidate in a set this small and are excluded from the MRR denominator. The figure therefore represents a selectively easier subpopulation rather than full-coverage performance. NDCG@10 at $K$=25, which includes the full user population, falls to 50.12%, well below the 53.76% achieved at $K$=125.

TABLE 5.4: Effect of candidate set size $K$ on NYC ($L$=2, %).

| $K$ | Recall@$K$ | HR@5 | HR@10 | NDCG@5 | NDCG@10 | MRR |
|---|---|---|---|---|---|---|
| 25 | 64.07 | 61.65 | 63.29 | 49.74 | 50.12 | 60.87 |
| 50 | 68.12 | 64.67 | 68.00 | 50.69 | 51.75 | 61.29 |
| 75 | 70.04 | 64.67 | 68.67 | 50.99 | 52.37 | 61.97 |
| 100 | 70.04 | 64.67 | 69.33 | 50.72 | 52.20 | 61.42 |
| **125** | **70.04** | **64.67** | **70.01** | **51.97** | **53.76** | **63.19** |
| 150 | 70.04 | 63.34 | 68.00 | 50.41 | 51.65 | 61.15 |
| 200 | 70.04 | 64.25 | 69.84 | 51.81 | 53.66 | 62.72 |

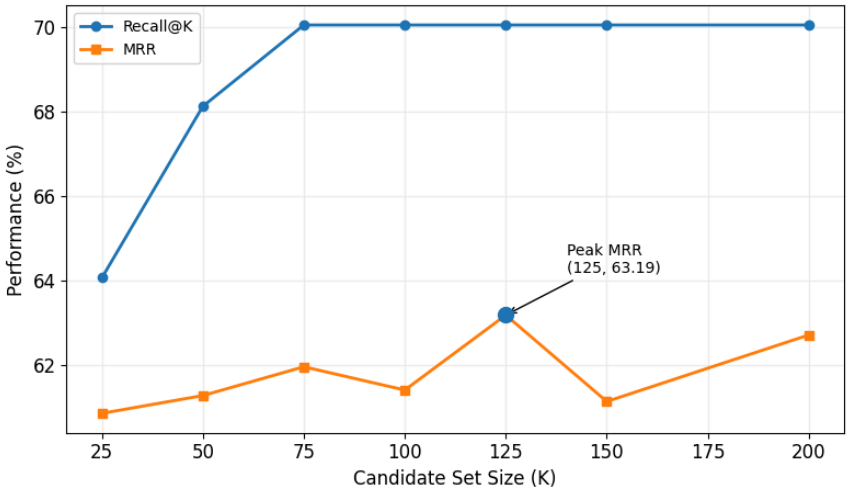


FIGURE 5.1: Recall@$K$ and MRR vs. candidate set size $K$ (NYC, $L$=2).

### 5.4.2 Transition Window Size

Table 5.5 shows that recall increases monotonically with $L$, while ranking quality peaks at $L$=2 and declines for larger windows. Averaging transition scores over more recent check-ins surfaces a broader set of candidates, but older transitions carry less directional signal about where the user is heading in the immediate term. As the window grows, the candidate set becomes more inclusive at the cost of introducing candidates whose relevance is harder for the Reranker to assess.

The $L$=4 row illustrates this tension precisely. It achieves the highest HR@5 in the table, 65.34%, recovering the most correct POIs into the top-5. Yet its MRR is the lowest at 59.60%, and NDCG@10 also falls relative to $L$=2, confirming that the broader candidate pool does not simply shift performance from top-1 to the wider list but actively degrades position-weighted quality. $L$=2 is the only configuration that simultaneously maximises

TABLE 5.5: Effect of transition window size $L$ on NYC ($K$=125, %).

| $L$ | Recall@$K$ | HR@5 | HR@10 | NDCG@5 | NDCG@10 | MRR |
|---|---|---|---|---|---|---|
| 1 | 68.02 | 63.94 | 67.15 | 51.61 | 52.76 | 62.16 |
| **2** | **70.04** | **64.67** | **70.01** | **51.97** | **53.76** | **63.19** |
| 3 | 70.04 | 64.67 | 68.00 | 50.93 | 51.96 | 61.21 |
| 4 | 72.06 | 65.34 | 67.33 | 50.58 | 51.19 | 59.60 |
| 5 | 74.19 | 64.01 | 65.33 | 50.58 | 50.98 | 60.73 |

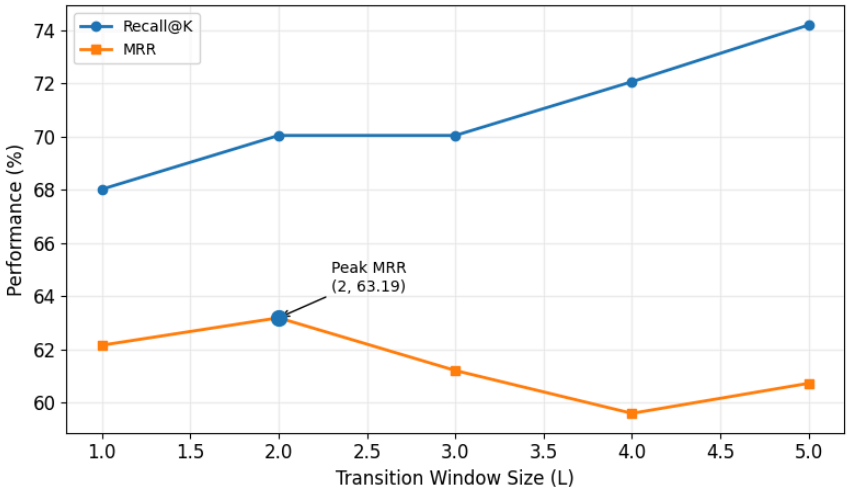


FIGURE 5.2: Recall@$K$ and MRR vs. transition window size $L$ (NYC, $K$=125).

NDCG@10 and MRR while also achieving recall saturation, making it the selection that best serves both ranking precision and position-weighted accuracy.

## 5.5 Discussion and Key Takeaways

Three analytical observations unify the results across all four experiments.

The consistent MRR-versus-HR asymmetry in the comparative results is a predictable consequence of what the cognitive adaptation loop is designed to do. Candidate retrieval is static; the episodic loop operates entirely on the reranking side. This means the loop's contribution should appear most strongly in metrics sensitive to rank ordering, and this is exactly what the results show across all three datasets. The asymmetry is not a limitation; it reflects the precise functional role of the episodic components within the architecture.

The ablation patterns confirm that memory and reflection serve fundamentally different functions rather than overlapping contributions to the same capability. Memory

produces proportional improvement across the full ranked list, indicating that it raises the general quality of the Reranker's contextual reasoning. Reflection produces concentrated improvement at the top-5 and top-1 positions, indicating that it specifically corrects the errors that keep the correct POI out of the highest ranks. The joint ablation demonstrates that these functions reinforce each other, with joint removal producing NDCG and MRR degradations far beyond what either individual removal predicts.

The backbone analysis establishes that the framework's advantage originates in the cognitive state design and not in any property of the specific model used. The performance degradation as backbone size decreases is real but moderate, and even the smallest tested configuration outperforms the strongest prior system by a clear margin. Within the range tested, instruction-following alignment quality explains more variance than raw parameter count, pointing toward alignment as the more important criterion when selecting a backbone for this type of framework.

# Chapter 6

# Conclusion and Future Work

## 6.1 Conclusion

This thesis introduced CogMaPOI, a five-agent LLM framework that treats next-POI recommendation as continual episodic inference rather than static single-pass prediction. The motivating claim is that human spatial decision-making draws on three distinct cognitive faculties simultaneously: stable long-term preferences accumulated over time, awareness of recent activity and short-term behavioural shifts, and the capacity to recognise and correct one's own reasoning errors. Every prior system in the literature, from deep sequential models and graph-based encoders to LLM-based prompt approaches and the most recent multi-agent frameworks, captures at most the first of these. CogMaPOI captures all three through three separately maintained state components assigned to dedicated agents within a strictly causal inference loop.

The first contribution is the formulation of next-POI prediction as continual episodic inference. Rather than producing a prediction from a fixed representation and terminating, the system maintains a cognitive state that evolves through natural-language updates as new observations accumulate. This formulation requires no gradient computation, no parameter modification, and no task-specific fine-tuning at any stage, making it applicable on top of any instruction-tuned language model.

The second contribution is the five-agent architecture that operationalises this formulation. The Profiler builds a structured identity prior from the historical trajectory, encoding stable preferences across six semantically meaningful fields. The Retrieval Agent assembles a user-specific candidate set by combining Markov transition

probabilities with personalised visit frequency, weighted adaptively by the user's exploration profile. The Memory Agent maintains a running summary of observed test-session behaviour, updated unconditionally after each episode. The Reflection Agent diagnoses prediction outcomes, reinforcing successful strategies and issuing corrective directives when predictions fail. The Reranker integrates all accumulated state at each episode to produce the final ranked output.

The third contribution is empirical validation across three Foursquare datasets and thirteen baselines spanning transformer-based, graph-based, generative, LLM-based, and multi-agent methods. CogMaPOI achieves relative MRR improvements of 56.88% on NYC, 46.36% on TKY, and 89.35% on CA over the strongest per-metric baselines. Ablation results confirm that memory and reflection make complementary, non-redundant contributions: memory enriches broad contextual reasoning while reflection specifically elevates top-rank precision. Even the smallest tested backbone, a 3B-parameter model, outperforms the strongest static baseline across all metrics, establishing that the performance advantage is architectural rather than a consequence of model scale or task-specific training.

## 6.2 Future Work

The results and architectural analysis identify several directions where this work can be meaningfully extended.

The retrieval stage currently relies on first-order Markov transition probabilities and historical visit frequency as its two statistical signals. Both capture patterns that are present in training data, but venues visited for reasons not reflected in sequential history, such as first-time discoveries, event-driven detours, or destinations reached through social recommendation, fall below the scoring threshold and remain unavailable to the Reranker. This stage currently recovers approximately 70% of ground-truth POIs, indicating a structural ceiling imposed by the retrieval design. Semantic similarity retrieval using dense representations of POI descriptions and user profiles could recover such venues by matching conceptual relevance rather than co-occurrence frequency. Geographic proximity retrieval could add spatially adjacent POIs that would not otherwise score highly. A two-stage pipeline combining fast statistical first-stage retrieval with semantic re-scoring as a second stage is a natural direction for improving candidate quality without abandoning the efficiency of the current design.

The evaluation window covers three episodes by design, following the CoMaPOI benchmark protocol to enable direct comparison with prior systems. This design does not answer whether the gains compound or plateau with more warm-up cycles. Running the episodic loop over five or ten episodes would clarify whether the reflection record continues to improve predictions as it accumulates more diagnoses, or whether it reaches a point of saturation once the primary failure modes for a given user have been identified and corrected. Longer sequences would also test the Memory Agent's ability to track sustained behavioural drift rather than short-term deviations.

The reflection mechanism currently operates within the boundaries of a single user's test session. Users with similar mobility profiles and exploration styles tend to trigger similar failure modes in the Reranker, which suggests a path toward cross-user knowledge transfer. Reflection patterns from users with well-characterised profiles could pre-populate the initial reflection state of new users with comparable histories, giving the system an informed corrective starting point rather than an empty record. This is most relevant for cold-start scenarios, where the historical trajectory is too sparse to build a reliable identity prior and the system must rely on borrowed inference from similar users.

Incorporating real-time contextual signals into the cognitive state is a further direction. Weather conditions, local events, and time-of-day context relative to the user's local timezone are known to influence mobility decisions, yet the current state representation contains none of these. Adding episode-level context of this kind to the Reranker prompt is structurally straightforward given the natural-language state design; the implementation challenge lies in sourcing reliable, timely, and consistent data across users and datasets rather than in the framework architecture itself.

# Bibliography


[1] W.-C. Kang and J. McAuley, "Self-attentive sequential recommendation," in *Proceedings of the 2018 IEEE International Conference on Data Mining (ICDM)*, 2018.

[2] F. Sun, J. Liu, J. Wu, C. Pei, X. Lin, W. Ou, and P. Jiang, "BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer," in *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM)*, 2019.

[3] Y. Dang, E. Yang, G. Guo, L. Jiang, X. Wang, X. Xu, Q. Sun, and H. Liu, "TiCoSeRec: Augmenting data to uniform sequences by time intervals for effective recommendation," *IEEE Transactions on Knowledge and Data Engineering*, 2024.

[4] R. Qiu, Z. Huang, H. Yin, and Z. Wang, "Contrastive learning for representation degeneration problem in sequential recommendation," in *Proceedings of the 15th ACM International Conference on Web Search and Data Mining (WSDM)*, 2022.

[5] S. Yang, J. Liu, and K. Zhao, "GETNext: Trajectory flow map enhanced transformer for next POI recommendation," in *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2022.

[6] H. Tao, J. Zeng, Z. Wang, M. Gao, and J. Wen, "Next POI recommendation based on spatial and temporal disentanglement representation," *IEEE International Conference on Web Services*, 2023.

[7] Y. Ye, L. Xia, and C. Huang, "Graph masked autoencoder for sequential recommendation," in *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2023.

[8] T. Huang, X. Pan, X. Cai, Y. Zhang, and X. Yuan, "Learning time slot preferences via mobility tree for next POI recommendation," in *Proceedings of the Thirty-Eighth*

*AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence*, 2024.

[9] Z. Li, A. Sun, and C. Li, "DiffuRec: A diffusion model for sequential recommendation," *ACM Transactions on Information Systems*, 2023.

[10] Y. Yang, S. Zhou, H. Weng, D. Wang, X. Zhang, D. Yu, and S. Deng, "Siamese learning based on graph differential equation for next-poi recommendation," *Applied Soft Computing*, 2024.

[11] P. Li, M. de Rijke, H. Xue, S. Ao, Y. Song, and F. D. Salim, "Large language models for next point-of-interest recommendation," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024.

[12] S. Feng, H. Lyu, F. Li, Z. Sun, and C. Chen, "Where to move next: Zero-shot generalization of LLMs for next POI recommendation," *IEEE Conference on Artificial Intelligence*, 2024.

[13] D. Wang, Y. Huang, S. Gao, Y. Wang, C. Huang, and S. Shang, "Generative next POI recommendation with semantic ID," in *KDD '25: Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2025.

[14] J. Chen, H. Wang, J. Shang, and Chaomurilige, "Interpretable embeddings for next point-of-interest recommendation via large language model question-answering," *Mathematics*, 2024.

[15] T. Wang and C. Wang, "Embracing large language models for point-of-interest recommendations," *IEEE Intelligent Systems*, 2024.

[16] ——, "Point-of-interest recommendations and large language models: A powerful combination," *IEEE Intelligent Systems*, 2025.

[17] T. Guo, X. Chen, Y. Wang, R. Chang, S. Pei, N. V. Chawla, O. Wiest, and X. Zhang, "Large language model based multi-agents: A survey of progress and challenges," in *Proceedings of the 33rd International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.

[18] Z. Xi *et al.*, "The rise and potential of large language model based agents: A survey," *Science China Information Sciences*, 2025.

[19] Z. Wang, Y. Yu, W. Zheng, W. Ma, and M. Zhang, "MACRec: A multi-agent collaboration framework for recommendation," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024.

[20] L. Zhong, L. Wang, X. Yang, and Q. Liao, "CoMaPOI: A collaborative multi-agent framework for next POI prediction bridging trajectory and language," in *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2025.

[21] S. Gao, Z. Wang, and Y. Zhao, "LMAPOI: Large language model based multi-agent framework for next POI recommendation," *IEEE International Symposium on Parallel and Distributed Processing with Applications (ISPA)*, 2025.

[22] D. Lopez-Paz and M. Ranzato, "Gradient episodic memory for continual learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[23] F. Sarfraz, E. Arani, and B. Zonooz, "Error sensitivity modulation based experience replay: Mitigating abrupt representation drift in continual learning," in *11th International Conference on Learning Representations (ICLR 2023)*. OpenReview, 2023.

[24] M. Boeltzig, N. Liedtke, S. Siestrup, F. Mecklenbrauck, M. F. Wurm, I. Bramão, and R. I. Schubotz, "The benefit of being wrong: How prediction error size guides the reshaping of episodic memories," *NeuroImage*, 2025.

[25] F. Pupillo, J. Ortiz-Tudela, R. Bruckner, and Y. L. Shing, "The effect of prediction error on episodic memory encoding is modulated by the outcome of the predictions," *npj Science of Learning*, 2023.

[26] G. Schillaci, U. Schmidt, and L. Miranda, "Prediction error-driven memory consolidation for continual learning," *KI: Künstliche Intelligenz*, 2020.